



HOME GARDEN TECHNOLOGY LEAFLET 17

SEED PRODUCTION

Seed production is a process with many stages, including research, breeding, production, certification, post-harvest handling and, if done commercially, marketing. For most home garden managers, commercial seeds are out of reach, because of their cost or because of limited access. Seed multiplication and saving, therefore, is an important practice for many rural households.

THE IMPORTANCE OF COMMUNITY-LEVEL SEED PRODUCTION AND THE CONSERVATION OF GENETIC RESOURCES

Genetic erosion is the loss of the genetic diversity of crops. The main cause of this, as reported by many countries worldwide, is the replacement of local varieties by "improved" exotic varieties and species. As more and more farmers introduce new commercial varieties, the number of varieties overall may be reduced.

The main cause of genetic erosion in Africa is the destruction of forests and bushland for large-scale commercial agriculture, to provide pastureland for livestock grazing and for the collection of fuelwood. In order to halt this process, many African countries have developed national plant genetic resources conservation programmes, many of which support on-farm conservation efforts. In Ethiopia, for example, farmers and plant breeders work together to restore plant varieties, such as those lost during the droughts of the 1980s.

For reasons relating to taste preference, cooking quality and storage characteristics, many rural African households and communities continue to cultivate and conserve local varieties as they have done for centuries. Seed production and selection by such households have contributed substantially to the maintenance of genetic diversity and have prevented the loss of some traditional crop varieties. Especially in risk-prone environments, farmers have overcome environmental constraints, such as diseases and harsh climates, by growing genetically diverse mixtures of crop varieties.

Home gardens in particular constitute a valuable part of the on-farm conservation of traditional varieties. Since women are often responsible for home gardens and the preparation of family meals, they play a large role in crop selection for the home garden. All home garden managers can play an even greater role in preserving the seeds of some of the uncultivated and semi-cultivated indigenous species, such as minor vegetables, fruits and other underused plant species that may not be part of national conservation and breeding programmes. It is also to their benefit when market outlets are available, because considerable income may be derived from the production and sale of seed from indigenous and exotic varieties of garden crops.

SELECTING SEED

When producing seed, the home garden manager must consider the following.

- Seeds should be saved from healthy, vigorous plants only.
- The plants from which seeds are to be saved should be marked and observed during their whole growth period.
- No seeds should be extracted from diseased and/or pest-infected plants. Undesirable plants should be removed and eaten by the household.
- If large amounts of seed are required, a plot should be reserved for seed production alone, in isolation from other similar crops.
- The exchange of good seeds among individual home garden managers facilitates the improvement of the seed stock over time and is, therefore, highly recommended.
- The local production of the seeds of cucurbits, chilli peppers and beans is of particular importance. Not only are they important food crops in Africa, they are also

especially liked for their colour, taste and spiciness.

HARVESTING AND PROCESSING SEED

Seeds are harvested and processed as follows:

- Ferment fleshy fruits until the seeds can be separated from the pulp. For some varieties of fruits, the pulp can be removed and eaten and the stones used as seed once properly dried.
- Once seeds are harvested, dry seeds in the shade. To avoid contamination with soil-borne diseases, do not place them on the ground. Once dry, seeds should be cleaned of all foreign matter before storage.
- Some seeds formed in pods should be left to dry in their pods on the plant. When the seeds are properly dried, the pods are removed from the plant. If the pods are removed from the plant before the seeds are properly dried, the pods must be dried further before they are placed in storage.

STORING SEED

Seeds should be stored in a cool, dry place, preferably in airtight containers (e.g. plastic bags or sealed clay pots). Generally, the moisture content of seeds should be kept low, especially if temperatures are high. Experts can provide advice to home garden managers on the viability of the different seed types.

